WENDIRAD DEMELASH TIKU

Adama, Oromia Region, Ethiopia wendiradame@gmail.com +251945224486 in/wendirad-demelash wendirad.com

SUMMARY

Senior Developer with extensive experience architect end-to-end AI and ML solutions using PyTorch, TensorFlow, and FastAPI. Expert in building advanced vision systems, generative models, and high-concurrency web scrapers. Combines deep technical proficiency in PostgreSQL and Docker with a researcher's mindset to deliver maintainable, high-performance software. Experienced in driving technical excellence through rigorous code reviews and modern DevOps practices.

EXPERIENCE

Python Developer

Turing .inc — **November 2022 - December 2025, USA (Remote)**

- Accelerated global LLM with more than 1.8 trillion parameters for coding proficiency by generating high-quality Python solutions and logical explanations to train the world's largest models.
- Improve model reasoning and reliability by resolving complex technical edge cases and creating detailed debugging documentation helping hundreds of millions of users.
- Conduct peer reviews of code and provide constructive feedback to maintain high development standards.

Academic and Research Assistant

Adama Science and Technology University — **October 2023 - July 2025, Adama, Ethiopia**

- Compiled and analyzed research data for faculty-led projects, contributing to the advancement of ongoing academic studies and publications.
- Facilitated academic success for undergraduates by providing targeted tutoring to bridge the gap between theoretical concepts and practical applications.
- Coordinated technical workshops and labs to enhance student proficiency through hands-on learning and structured curriculum support.

Chief Technology Officer (CTO)

Backos Technology — **September 2020 - January 2024, Adama, Ethiopia**

- Aligned technology with business objectives by directing long-term technical strategies to foster company growth and innovation.
- Scaled organizational output by building and leading high-performing technical teams through collaborative excellence and mentorship.
- Enhanced product marketability by overseeing the end-to-end development of cutting-edge technology solutions and infrastructure.

Full Stack Web Developer

Adama Science and Technology University, CoEEC — **May 2022 - October 2022, Adama, Ethiopia**

- Digitized inventory management for 3 university departments by leading the development of a web-based system to streamline operational efficiency.
- Automated tracking for 1,000+ inventory items by implementing rigorous stock balance verification and real-time report analytics.
- Improved administrative workflows through the design of a user-friendly interface tailored for high-volume data management and accurate reporting for 100+ staff members.

Full Stack Web Developer

Divergent Technology — **February 2022 - July 2022, Adama, Ethiopia**

- Architect an end-to-end web solution for AI and ML education by designing the system architecture and implementing dynamic content managers for courses and student registration.
- Developed a custom, high-fidelity front-end to deliver an administrative dashboard and user interface aligned with explicit client design specifications.
- Streamlined product delivery by managing the full lifecycle through iterative development.

Fullstack Developer

Backos Technology — **January 2020 - August 2020, Adama, Ethiopia**

- Engineered robust, scalable software solutions by integrating front-end and back-end technologies to support high-growth business requirements.
- Delivered user-centric digital products by collaborating with designers and product managers to translate user needs into functional technical specifications.
- Streamlined full-stack development cycles using Python and modern web frameworks to ensure the rapid delivery of maintainable and efficient codebases.

PROJECT

FidelGen: Visual Amharic Text Generation Using a Multi-Conditional Diffusion Model

MSc Thesis Project at Adama Science and Technology University • September 2024 - June 2025

- Developed a diffusion-based text-to-image model for Amharic script using parallel cross-attention to bridge the language-visual gap.
- Enhanced text accuracy by implementing a dual-loss training approach, combining pixel and OCR losses with a progressive training strategy.
- Validated high-fidelity rendering and consistency through quantitative analysis using CLIP Score and Character Error Rate.

FidelGen OCR

MSc Thesis Project at Adama Science and Technology University • September 2024 - June 2025

- An Amharic OCR system trained on a combination of HUST and synthetic data, designed to recognize characters and words with high accuracy across varied styles and visual conditions.

GenSyn

MSc Thesis Project at Adama Science and Technology University • December 2024 - June 2025

- Engineered a high-volume data generation tool in Python to produce synthetic text-on-image datasets for training OCR and computer vision models.
- Automated the creation of diverse training samples by implementing randomized text overlays, dynamic background coloring, and varied typographic structures.

Drought Prediction Using Hybrid CNN-LSTM

MSc Thesis Project at Adama Science and Technology University • February 2024 - August 2024

- A Python-based drought prediction model combining CNN and LSTM for temporal modeling on preprocessed climate data to accurately forecast drought occurrences in Ethiopia regions.

Rebuni Media Monitoring System

BSc Graduation Project at Adama Science and Technology University • September 2022 - July 2023

- Developed a comprehensive system for analyzing social media and broadcast media engagements, supporting strategic decision-making through advanced data analysis.

EDUCATION

M.Sc in Computer Science and Engineering (Computer Vision)

Adama Science and Technology University • Ethiopia • 2023 - 2025 • 4/4

- Write a masters thesis titled "*FidelGen: Visual Amharic Text Generation Using a Multi-Conditional Diffusion Model*".

B.Sc Dual major in Computer Science and Engineering & Electronics and Communication Engineering

Adama Science and Technology University • Ethiopia • 2018 - 2023 • 3.6/4

- A merit-based, selective dual-major track requiring mastery of two full curricula while maintaining a consistently high GPA.

SKILLS

Technical Skills and Tools

Python, JavaScript, PyTorch, TensorFlow, Data Analysis, Data Visualization, Django, PostgreSQL, MongoDB, Git, MySQL, FastAPI, Docker, Visual Studio Code, PyCharm, Jupyter Notebook

Soft Skills

Teamwork, Effective Communication, Problem-Solving, Leadership, Adaptability, Discipline, Collaboration, Time Management, Attention to Detail, Proactive Approach

LANGUAGES

English

C1 (Advanced)

Amharic

Native

AWARDS & HONORS

Zindi Certificate, Data Science

Zindi • 2022

- Earned for significant contributions to data science projects on Zindi, demonstrating expertise in data analysis and machine learning algorithms.

REFERENCES

Dr. Worku Jifara

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Dr. Teklu Urgessa

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